Effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification using the SomaScan and Olink assays

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Abstract

SomaScan and Olink are the leading aptamer- and antibody-based platforms in current proteomics research, each of them capable of measuring thousands of human proteins with a broad range of endogenous concentrations. In this study, we investigated the effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification across 10,776 (11K SomaScan) and 1,472 (Olink Explore 1536) analytes, respectively. Using SomaScan, we found two distinct groups, each one consisting of 4% of all aptamers, affected by either hemolysis or freeze-thaw cycles. Using Olink, probes impacted by freeze-thaw cycles were in line with the observations from SomaScan, with 6% of analytes affected. In dramatic contrast, however, we found that nearly half of all measured Olink probes were significantly impacted by hemolysis. Moreover, by a variety of statistical approaches, we observed that Olink probes affected by hemolysis target proteins with a larger number of annotated protein-protein interactions. Given the extent of the observed nuisance effects, we propose that unbiased, quantitative methods of evaluating hemolysis, such as the hemolysis index successfully implemented in many clinical laboratories, should be adopted in proteomics studies, particularly in those using the Olink assay. We provide detailed results for each SomaScan and Olink probe in the form of extensive Supplementary Data files to be used as resources for the growing user communities of both platforms.

Introduction

It is well known that pre-analytical variation, defined as differences in collection, handling, storage, and sample processing, can affect the measurement of protein abundances, therefore impacting the scientific discovery potential of proteomics assays. On the one hand, pre-analytical variation may manifest itself in the form of in vitro or spurious hemolysis.[1–4] Indeed, estimated to range between 1 and 5% of the total samples processed in clinical settings, hemolyzed specimens are reportedly the most prevalent manifestation of procedural mistakes in laboratory testing.[5] On the other hand, pre-analytical variation may occur also as a result of repeated freeze-thaw cycles. Typically, blood samples are stored in -80 °C freezers prior to analysis; in practice, however, biobank samples may undergo multiple cycles of aliquot thawing and refreezing. Previous studies have investigated the effects of repeated freeze-thaw cycles in a variety of systems, e.g. lipoproteins and metabolites measured by nuclear magnetic resonance,[6] plasma proteomics measured by mass spectrometry,[7] and several others.[8–10] In this context, the aim of our study was to explore the concurrent effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification using SomaScan[11] and Olink,[12] which are, respectively, the leading aptamer- and antibody-based platforms in current proteomics research.

SomaScan[11] is a highly multiplexed, aptamer-based assay capable of simultaneously measuring thousands of human proteins broadly ranging from femto- to micro-molar concentrations. This technology relies on protein-capture SOMAmer (*Slow Offrate Modified Aptamer*) reagents,[13] designed to optimize high affinity, slow off-rate, and high specificity to target proteins. These targets extensively cover major molecular functions including receptors, kinases, growth factors, and hormones, and span a diverse collection of secreted, intracellular, and extracellular proteins or domains. In this study, we utilized the 11K (v5.0) SomaScan assay, which is the most recent version of this platform, capable of measuring 10,776 different SOMAmers targeting human proteins.[14]

Olink,[12] in turn, implements the so-called Proximity Extension Assay (PEA) technology, a dual-recognition immunoassay in which two matched antibodies labeled with unique DNA oligonucleotides simultaneously bind to a target protein in solution. By bringing the two antibodies into proximity, their DNA oligonucleotides are allowed to hybridize, serving as a DNA barcode to identify and quantify the abundance of a specific target antigen. In this study, we utilized the Olink Explore assay to measure 1,472 analytes across four panels focused on different disease and biological processes, namely cardiometabolic, inflammation, neurology, and oncology.

Several studies have previously compared SomaScan and Olink across a variety of different technical aspects.[15–21] To the best of our knowledge, however, no studies to date have been dedicated to investigate the impact of pre-analytical effects on protein abundance quantification measurements obtained with either platform. As anticipated, in this work we explored the concurrent effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification using SomaScan and Olink. To that end, we analyzed a total of 90 plasma samples from 15 healthy adult human donors evenly distributed in age and sex under different hemolysis and freeze-thaw conditions. By implementing mixed-effects models, we identified analytes showing significant effects resulting from pre-analytical perturbations. Furthermore, we mapped all combinations of SomaScan and Olink probes to unique UniProt IDs and investigated possible associations with protein characteristics obtained from the UniProt Knowledgebase. Detailed results are provided in the form of Supplementary Data files to be used by the SomaScan and Olink research communities as technical resource materials.

Experimental Section

Study design

Fig. 1 shows details of the study design. Blood samples were collected from 15 adult participants enrolled in the Baltimore Longitudinal Study of Aging[22] (BLSA), a study of normative human aging established in 1958 and conducted by the National Institute on Aging, NIH. The study protocol was conducted in accordance with Declaration of Helsinki principles and was reviewed and approved by the Institutional Review Board of the NIH's Intramural Research Program. Written informed consent was obtained from all participants. Samples used in this study were obtained from recent visits scheduled between February and April 2023. The participants' sex and age were approximately evenly distributed, namely: 7 females (47%) with ages in the 24-89 years old range (mean: 54.6 y.o., median: 51 y.o.) and 8 males (53%) with ages in the 25-95 years old range (mean: 58.9 y.o., median: 56.5 y.o.). Each biological sample was split into six technical replicates obtained from combining the hemolyzed/non-hemolyzed condition with 3, 10 and 20 freeze-thaw cycles, which resulted in a total of 90 samples. The resulting samples were then analyzed with the SomaScan and Olink proteomics platforms. Because of calibrators, quality control and buffer wells, a single SomaScan plate is only able to accommodate up to 85 user-provided samples; we thus randomly selected 5 samples to be removed from the SomaScan study. Olink, on the other hand, was able to fit all 90 samples in a single plate (after removing two sample controls).

Sample preparation

From each participant, fasting whole blood was collected by venipuncture into two (3.0 mL K2 EDTA BD Vacutainer) collection tubes. Hemolysis was induced in one of the collection tubes by forcing blood through 28G needles 3 to 4 times. Samples were visually inspected to ensure they had a qualitatively similar level of hemolysis. The blood was centrifuged at 2300 rpm at 20 °C for 15 minutes to obtain plasma. Three 500 μL aliquots of plasma were obtained from each tube and stored at -80 °C until assayed. For the freezing/thawing cycles, each aliquot was put in dry ice for 15 minutes and subsequently left at room temperature until thawed and additional 10 minutes after complete thawing. This procedure was performed 3, 10, or 20 times depending on the condition.

The SomaScan assay

Proteins were measured using the 11K (v5.0) SomaScan assay (SomaLogic, Inc.; Boulder, CO, USA). This technology relies on aptamers (SOMAmers), which are single-stranded, chemically-modified nucleic acids, selected via the so-called SELEX (Systematic Evolution of Ligands by EXponential enrichment) process, designed to optimize high affinity, slow off-rate, and high specificity to target proteins. The 11K assay consists of 10,776 SOMAmers targeting annotated human proteins. In order to cover a broad range of endogenous concentrations, SOMAmers are binned into different dilution groups, namely 20% (1:5) dilution for proteins typically observed in the femto- to pico-molar range (which comprise 80% of all human protein SOMAmers in the assay), 0.5% (1:200) dilution for proteins typically present in nano-molar concentrations (18% of human protein SOMAmers in the assay), and 0.005% (1:20,000) dilution for proteins in micro-molar concentrations (2% of human protein SOMAmers in the assay). The human plasma volume required is 55 μL per sample. The experimental procedure follows a sequence of steps, namely: (1) SOMAmers are synthesized with a fluorophore, photocleavable linker, and biotin; (2) diluted samples are incubated with dilution-specific SOMAmers bound to streptavidin beads; (3) unbound proteins are washed away, and bound proteins are tagged with biotin; (4) UV light breaks the photocleavable linker, releasing complexes back into solution; (5) non-specific complexes dissociate while specific complexes remain bound; (6) a polyanionic competitor is added to prevent rebinding of non-specific complexes; (7) biotinylated proteins (and bound SOMAmers) are captured on new streptavidin beads; and (8) after SOMAmers are released from the complexes by denaturing the proteins, fluorophores are measured following hybridization to complementary sequences on a microarray chip. The fluorescence intensity detected on the microarray, measured in *RFU* (*Relative Fluorescence Units*), is assumed to reflect the amount of available epitope in the original sample. In order to account for intra- and inter-plate effects, a normalization procedure is applied, consisting of a sequence of steps, whose main elements are hybridization normalization, median signal normalization, plate-scale normalization, and inter-plate calibration.[23,24] Results in this paper are based on the fully normalized data. For downstream analyses, protein relative concentration was assumed as given by $\log_{10}(RFU)$. Further information about the SomaScan assay is available on the manufacturer's website (www.somalogic.com).

The Olink assay

Proteins were measured using Olink Explore 1536 (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. We used four panels focused on different disease and biological processes, namely Cardiometabolic (369 analytes), Inflammation (368 analytes), Neurology (367 analytes), and Oncology (368 analytes), for a total of 1,472 analytes. For each panel, the human plasma volume required is 2.8 μL per sample. The technology behind the Olink protocol is based on the so-called Proximity Extension Assay (PEA),[12] which is coupled with readout via next-generation sequencing. In brief, pairs of oligonucleotide-labeled antibody probes designed for each protein bind to their target, bringing the complementary oligonucleotides in close proximity and allowing for their hybridization. The addition of a DNA polymerase leads to the extension of the hybridized oligonucleotides, generating a unique protein identification "barcode". Next, library preparation adds sample identification indices and the required nucleotides for Illumina sequencing. Prior to sequencing using the Illumina NovaSeq 6000, libraries go through a bead-based purification step and the quality is assessed using the Agilent 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA, USA). The raw count data was generated using bcl2counts (v2.2.0) and was quality controlled, normalized and converted into Normalized Protein eXpression (NPX), Olink's proprietary unit of relative abundance. Data normalization is performed using an internal extension control and an external plate control to adjust for intra- and inter-run variation. All assay validation data (detection limits, intra- and inter-assay precision data, predefined values, etc.) are available on the manufacturer's website (www.olink.com).

Protein annotations

SOMAmers are uniquely identified by their aptamer sequence ID ("SeqId"), but the relationship between SOMAmers and annotated proteins is not one-to-one. In the 11K assay, the 10,776 SOMAmers that target human proteins are mapped to 9,609 unique UniProt IDs, yielding a total of 10,893 unique SOMAmer-UniProt ID pairs. Supplementary Data 1 provides the list of SOMAmer-UniProt ID pairs, along with protein annotations extracted from the UniProt Knowledgebase (UniProtKB), including length (number of amino acids in the canonical sequence), mass (molecular weight), pH dependence, Redox potential, temperature dependence, protein interactions, and subcellular location. Analogously, the 1,472 Olink probes are mapped to 1,465 unique UniProt IDs, yielding 1,476 unique Olink-UniProt ID pairs, which are listed in Supplementary Data 2 along with UniProtKB annotations. The overlap between the 11K SomaScan and the four Olink panels used in our study comprises 1,872 SOMAmers and 1,353 Olink probes mapped to 1,356 unique UniProt IDs, yielding a total of 1,893 SOMAmer/Olink/UniProt ID triplets, listed in Supplementary Data 3. Notice that the list comprises 1,891 unique SOMAmer/Olink pairs; the "Duplicated" column flags the two duplicated SOMAmer/Olink pairs. Below we describe an analysis to find SOMAmer/Olink pairs that are affected by hemolysis and freeze-thaw cycles in both platforms (see Fig. 4); results are provided in the "Paired signif." column, indicating also the direction of change (for instance, "H: O+S-" indicates a pair that is affected by hemolysis, such that the Olink probe is increased and the paired SOMAmer decreased in hemolyzed samples).

Statistical methods

For each probe in the SomaScan and Olink assays, mixed-effects models were run according to the formula:

$$concentration \sim (1|subject) + (1|sex) + (1|age) + H + FTC \ , \tag{1}$$

where hemolysis (H) and freeze-thaw cycles (FTC) were treated as fixed effects, and subject ID, sex and age were considered random effects. Age was binarized using tertile bins; hemolysis was coded as $H = 1$ for hemolyzed samples and $H = 0$ for non-hemolyzed samples, and the freeze-thaw cycle variable was defined as $FTC = 1$, 2, and 3 corresponding to the three conditions performed (3, 10, and 20 cycles, respectively). Mixed-effects models were implemented in a script via the lmer function from R package lme4 v.1.1.35.5 with the argument REML=FALSE to optimize the log-likelihood.[25] The statistical significance of effects associated to hemolysis was assessed by performing a likelihood ratio test to compare the full model from Eq. (1) against the null model:

$$concentration \sim (1|subject) + (1|sex) + (1|age) + FTC \ , \tag{2}$$

using the anova function from R package stats v.4.4.0. Similarly, the significance of effects associated with freeze-thaw cycles was derived from comparing the full model against the null model:

$$concentration \sim (1|subject) + (1|sex) + (1|age) + H \ . \tag{3}$$

Data and Software Availability

Data analysis was performed in R v.4.4.0. Protein annotations from UniProtKB were extracted using the R package queryup[26] v.1.0.5. Mixed-effects models were implemented via lme4 v.1.1.35.5. Plots were generated using R packages RColorBrewer v.1.1-3, calibrate v.1.7.7, and VennDiagram v.1.7.3. BioRender was used to prepare Fig. 1. Anonymized datasets and custom R source code used in our analyses are available on the Open Science Framework repository [NOTE: to be made publicly available upon publication].

Results and Discussion

Fig. 2 displays Volcano plots showing significantly affected probes, obtained from mixed-effects models separately run for SomaScan (top panels, a-b) and Olink (bottom panels, c-d), as indicated. Left-side panels (a,c) show the effects of hemolysis, while right-side panels (b,d) show the effects of repeated freeze-thaw cycles. Detailed results are provided for SomaScan (Supplementary Data 4) and Olink (Supplementary Data 5), respectively. For SomaScan, we found 387 (4%) SOMAmers with significant effects associated to hemolysis (Fig. 2(a)), approximately evenly distributed in the positive (210 SOMAmers) and negative (177 SOMAmers) directions. A similar picture emerged when considering freeze-thaw cycle effects: we found 466 (4%) SOMAmers affected (Fig. 2(b)), although more biased towards the negative direction (301 SOMAmers) relative to the positive one (165 SOMAmers). For Olink, in striking contrast, we found nearly half of the probes significantly affected by hemolysis (Fig. 2(c)). Indeed, we observed 718 (49%) affected analytes, most of which appeared to increase in hemolyzed samples (624 in the positive direction vs 94 in the negative direction). The effects of freeze-thaw cycles (Fig. 2(d)) were much less dramatic and more in line with the observations from SomaScan; we found 84 (6%) analytes affected and biased in a similar $\sim 2:1$ ratio towards the negative direction (58 analytes) relative to the positive one (26 analytes).

Table 1: **Olink probes significantly affected by hemolysis and freeze-thaw cycles.**

	Hemolysis		**Freeze-thaw cycles**	
Panel	**n.s.**	**significant**	**n.s.**	**significant**
Cardiometabolic	225 (61%)	144 (39%)	349 (95%)	20 (5%)
Inflammation	192 (52%)	176 (48%)	350 (95%)	18 (5%)
Neurology	170 (46%)	197 (54%)	342 (93%)	25 (7%)
Oncology	167 (45%)	201 (55%)	347 (94%)	21 (6%)

For the sake of illustration, Fig. 3 shows protein abundance trajectories from the Olink assay. In particular, panels (a-b) show PPME1 (protein phosphatase methylesterase 1; Olink ID 21347) across freeze-thaw cycles for hemolyzed (panel (a)) and non-hemolyzed (panel (b)) samples. PPME1 is the protein with strongest hemolysis effects in the Olink platform ($\beta_H = 4.3,\ \mathrm{p-value} = 1.2 \times 10^{-62}$) and it also displays mild but significant freeze-thaw cycle effects ($\beta_{FTC} = 0.13,\ \mathrm{p-value} = 0.013$), both in the positive direction (i.e. increasing protein concentration with both hemolysis and repeated freeze-thaw cycles). Similarly, panels (c-d) show KLK13 (kallikrein related peptidase 13; Olink ID 21291), which is the protein with strongest freeze-thaw cycle effects ($\beta_{FTC} = -0.14,\ \mathrm{p-value} = 5.4 \times 10^{-9}$). KLK13's concentration decreases with increasing freeze-thaw cycles and shows no significant associations with hemolysis ($\mathrm{p-value} = 0.71$). Both PPME1 (OID21347) and KLK13 (OID21291) belong to the oncology panel, an observation that may suggest that pre-analytical variation effects may be associated with the panels Olink probes belong to. Table 1 shows the number of probes in each panel affected by hemolysis and freeze-thaw cycles; we did indeed observe a significant association in the former (Fisher's exact test $\mathrm{p-value} = 4.4 \times 10^{-5}$), showing an enrichment of affected probes in the oncology and neurology panels, whereas freeze-thaw cycle effects did not appear to be significantly associated with panel of origin ($\mathrm{p-value} = 0.72$).

Fig. 4 shows Venn diagrams displaying the number of probes significantly affected by hemolysis and freeze-thaw cycles. For Olink (Fig. 4(a)), 42 (2.8%) probes are affected by both, which is a non-significant overlap (Fisher's exact test $\mathrm{p-value} = 0.82$). For SomaScan

(Fig. 4(b)), only 14 (0.13%) SOMAmers are affected by both, which is also a non-significant overlap (Fisher's exact test $\mathrm{p-value} = 0.61$). In order to compare across platforms, we first needed to map Olink probes and SOMAmers to UniProt targets, then consider those UniProt targets in common to associate Olink probes to SOMAmers. This procedure yielded a total of 1,893 SOMAmer/Olink/UniProt ID triplets and 1,891 unique SOMAmer/Olink ID pairs, listed in Supplementary Data 3. We found only 30 SOMAmer/Olink pairs affected by hemolysis in both platforms; of these, only 19 agree in the direction of change (17 increase, 2 decrease abundance in hemolyzed samples), whereas the remaining 11 change in opposite directions. For freeze-thaw cycles, we only found 5 SOMAmer/Olink pairs affected, and only one of them agrees in the direction of change.

Given the large proportion of Olink probes affected by hemolysis, we investigated possible associations with protein characteristics obtained from UniProtKB (see Supplementary Table 2). We found that the number of protein interactions is larger for protein probes affected by hemolysis compared with protein probes not significantly affected by hemolysis. Calculated over all 1,476 Olink/UniProt ID pairs, we found that the median number of interacting proteins is 5 for affected Olink probes and 2 for non-affected probes (Wilcoxon test $\mathrm{p-value} = 3.5 \times 10^{-18}$). Removing 343 (23%) pairs that show no interacting proteins (possibly due to limitations in the available annotation database), the observed association persists, with the median number of interacting proteins equal to 6 for affected Olink probes and equal to 4 for non-affected probes (Wilcoxon test $\mathrm{p-value} = 1.2 \times 10^{-11}$). Fig. 5(a) shows the percentile distribution comparison of the number of interacting proteins for Olink targets not significantly affected by hemolysis (blue) versus those significantly affected (red), which confirm the observed association. As a complementary analysis, we explored the association between the number of interacting proteins and the absolute value of beta coefficients ($|\beta_H|$) obtained from the mixed effects models. We found Spearman's correlation to be significant ($r = 0.23$ and $\mathrm{p-value} = 7.1 \times 10^{-15}$), with the caveat that this method cannot compute exact p-values with ties. Alternatively, taking logs to account for the fact that both variables have long-tailed distributions, Pearson's correlation was found to be similarly significant ($r = 0.22$ and $\mathrm{p-value} = 4.6 \times 10^{-14}$). Finally, by binning the number of interacting proteins among $n_b = 5$ bins of similar size, we regressed $\log(|\beta_H|)$ against the bin index $i_b = 1, \ldots, n_b$ and found a positive, significant regression slope ($\mathrm{p-value} = 5.9 \times 10^{-14}$). The binned data with the overlaid linear fit is shown in Fig. 5(b). Hemolysis-sensitive SOMAmers, in contrast, were not found to be associated with the number of interacting proteins.

Conclusions

We investigated the effects of in vitro hemolysis and repeated freeze-thaw cycles in protein abundance quantification using the SomaScan and Olink assays, the leading aptamer- and antibody-based platforms in current proteomics research, respectively. For SomaScan, we found two distinct groups, each one consisting of 4% of all measured SOMAmers, affected by either hemolysis or freeze-thaw cycles. For Olink, probes affected by freeze-thaw cycles were in line with the observations from SomaScan, with 6% of probes affected and biased in a similar $\sim 2:1$ ratio towards the negative direction (i.e. decreasing abundance by increasing freeze-thaw cycles) relative to the positive one. In striking contrast, however, we found that nearly half of the Olink probes were significantly affected by hemolysis and most of them biased towards the positive direction (i.e. increasing abundance by hemolysis), whereas the much smaller proportion of affected SOMAmers appeared approximately evenly distributed in both directions. Due to the large proportion of Olink probes affected by hemolysis, we investigated possible associations with protein characteristics obtained from UniProtKB and found a significant association between hemolysis-sensitive Olink probes and the number of annotated protein-protein interactions of their targets. While the Olink analysis revealed a large number of protein measurements affected by significant in vitro hemolysis, this is not completely unexpected, as immunoassays of heavily hemolyzed samples have shown evidence of interference[27,28] and a similar observation was made in a human plasma biomarker study using a similar platform.[29] Although we may conjecture that the observed differences between SomaScan and Olink may reflect differences in the detection mechanisms at play at the molecular scale between these two complementary technologies (namely, modified DNA aptamer-based vs antibody-based), this is a topic that certainly deserves further investigation and lies beyond the scope of the present work.

Our observations highlight the need to adopt an extremely careful approach for sample pre-processing and handling, in order to ensure the integrity of proteomics measurements and downstream analysis, in particular for studies based on Olink measurements. In clinical practice, the phenomenon of in vitro hemolysis affecting clinical samples is well known. Historically, hemolysis was detected by visual inspection of the specimen after centrifugation, but a newer approach is based on the so-called hemolysis index (HI).[30,31] Most clinical chemistry platforms are nowadays equipped with automated HI detection systems, which involve the rapid spectrophotometric measurement of free hemoglobin in serum or plasma along with decision rules for the systematic handling of specimens based on tolerance range values. We suggest that, upon further investigation, HI-based scores integrated with assay- and probe-specific suitable thresholds may provide an unbiased, quantitative procedure to flag and, even, correct proteomics measurements that may be affected by hemolysis. To a lesser extent, quantitative metrics to assess freeze-thaw cycles may be also implemented for similar purposes. In this regard, SomaScan has recently begun to explore and implement post-hoc, machine-learning-based estimates of pre-analytical variation called SomaSignal Tests, which aim at quantifying a variety of sample processing factors, including fed-fasted time, number of freeze-thaw cycles, time-to-decant, time-to-spin, and time-to-freeze.[32]

Accompanying this paper, detailed results for each Olink and SOMAmer probe are provided in the form of extensive Supplementary Data files. As SomaScan and Olink become more widely adopted and utilized as state-of-the-art tools for proteomics discovery, we hope that our work will serve as a valuable technical reference and resource for the growing user communities of both platforms.

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Supporting Information Available

The following files are available online:

- Supplementary Data 1: List of 10,893 unique SOMAmer-UniProt ID pairs, along with protein annotations extracted from UniProtKB.
- Supplementary Data 2: List of 1,476 unique Olink-UniProt ID pairs, along with protein annotations extracted from UniProtKB.
- Supplementary Data 3: List of 1,893 unique SOMAmer-Olink-UniProt ID triplets, along with protein data extracted from UniProtKB.
- Supplementary Data 4: Mixed-effects model results to assess the effect of hemolysis and freeze-thaw cycles on SomaScan's protein concentration measurements.
- Supplementary Data 5: Mixed-effects model results to assess the effect of hemolysis and freeze-thaw cycles on Olink's protein concentration measurements.

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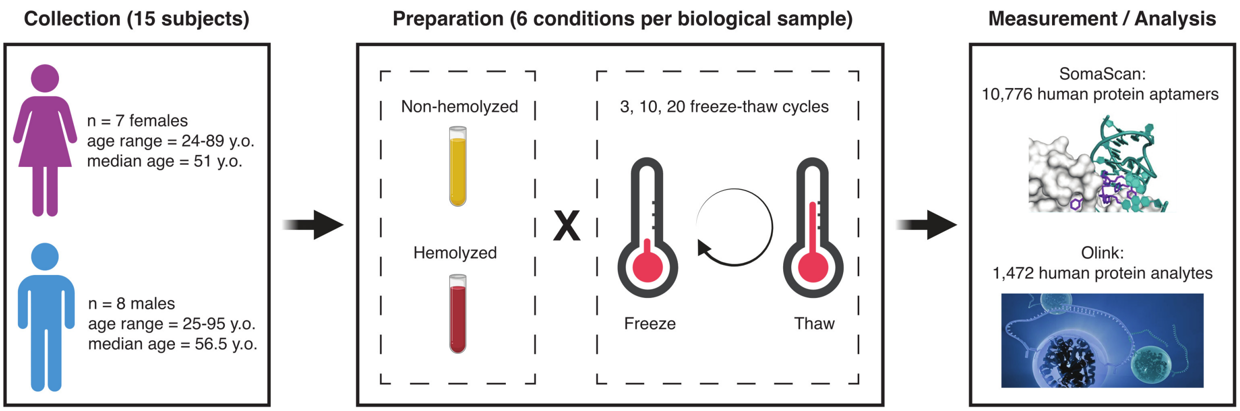

Figure 1: **Study design.** Blood samples were collected from 15 participants in the Baltimore Longitudinal Study of Aging (BLSA), approximately evenly distributed across sex and age. Each biological sample was split into six technical replicates obtained from combining the hemolyzed/non-hemolyzed condition with 3, 10 and 20 freeze-thaw cycles. The resulting samples were then analyzed with the SomaScan and Olink proteomics platforms.

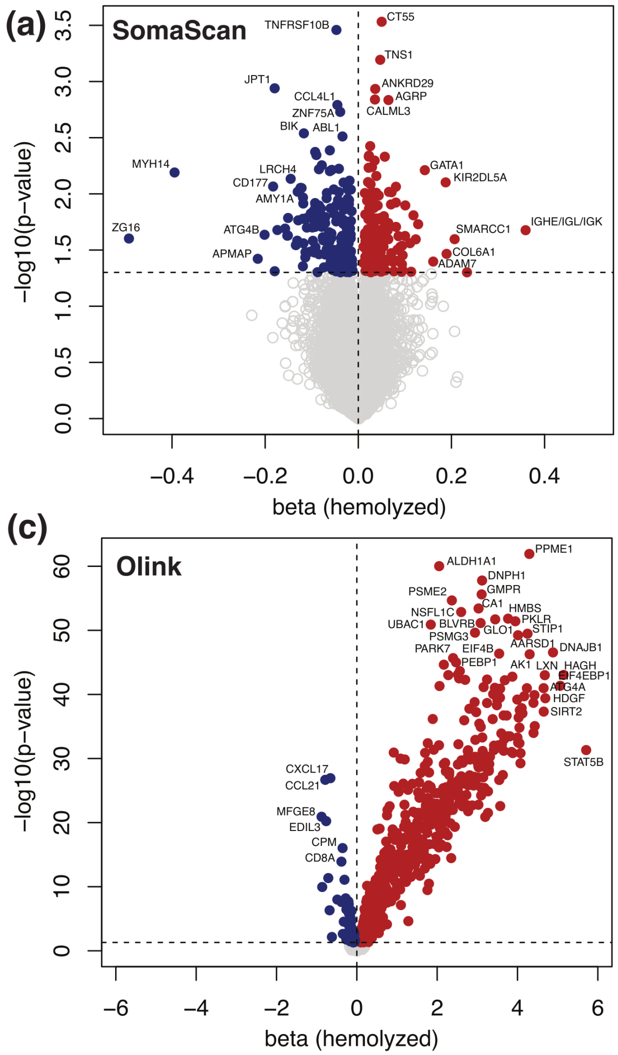

Figure 2: **Volcano plots showing significantly affected probes**. Mixed-effects models were separately run for SomaScan **(a-b)** and Olink **(c-d)**, as indicated. Left-side panels **(a,c)** show the effects of hemolysis, while right-side panels **(b,d)** show the effects of repeated freeze-thaw cycles. Significance is shown on the y-axis as unadjusted $-\log_{10}(\mathrm{p-value})$. The horizontal dashed lines correspond to the $\mathrm{p-value} = 0.05$ threshold.

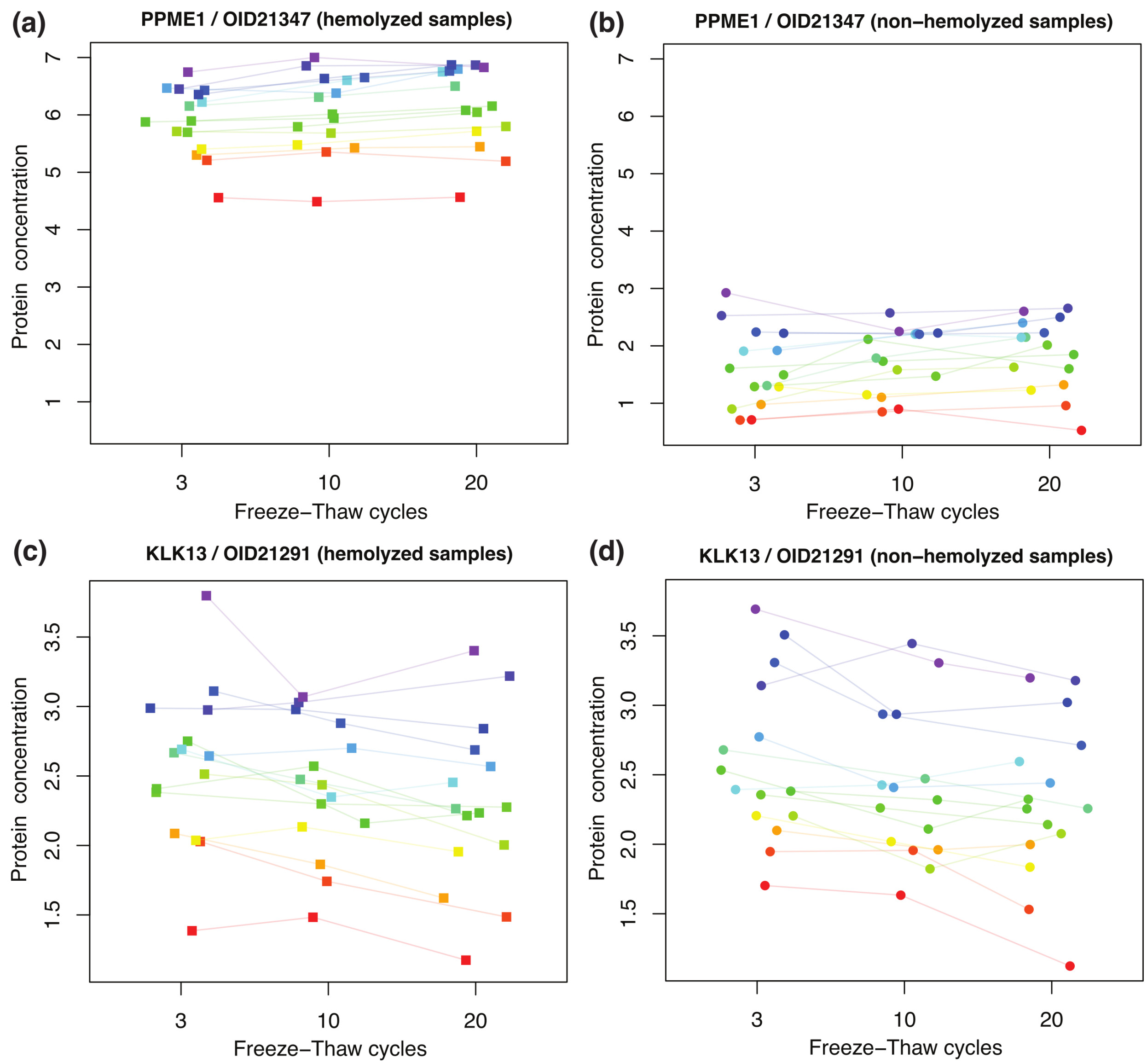

Figure 3: **Protein abundance trajectories from the Olink assay. (a-b)** PPME1 (protein phosphatase methylesterase 1) across freeze-thaw cycles for hemolyzed (panel **(a)**) and non-hemolyzed (panel **(b)**) samples. **(c-d)** KLK13 (kallikrein related peptidase 13) across freeze-thaw cycles for hemolyzed (panel **(c)**) and non-hemolyzed (panel **(d)**) samples.

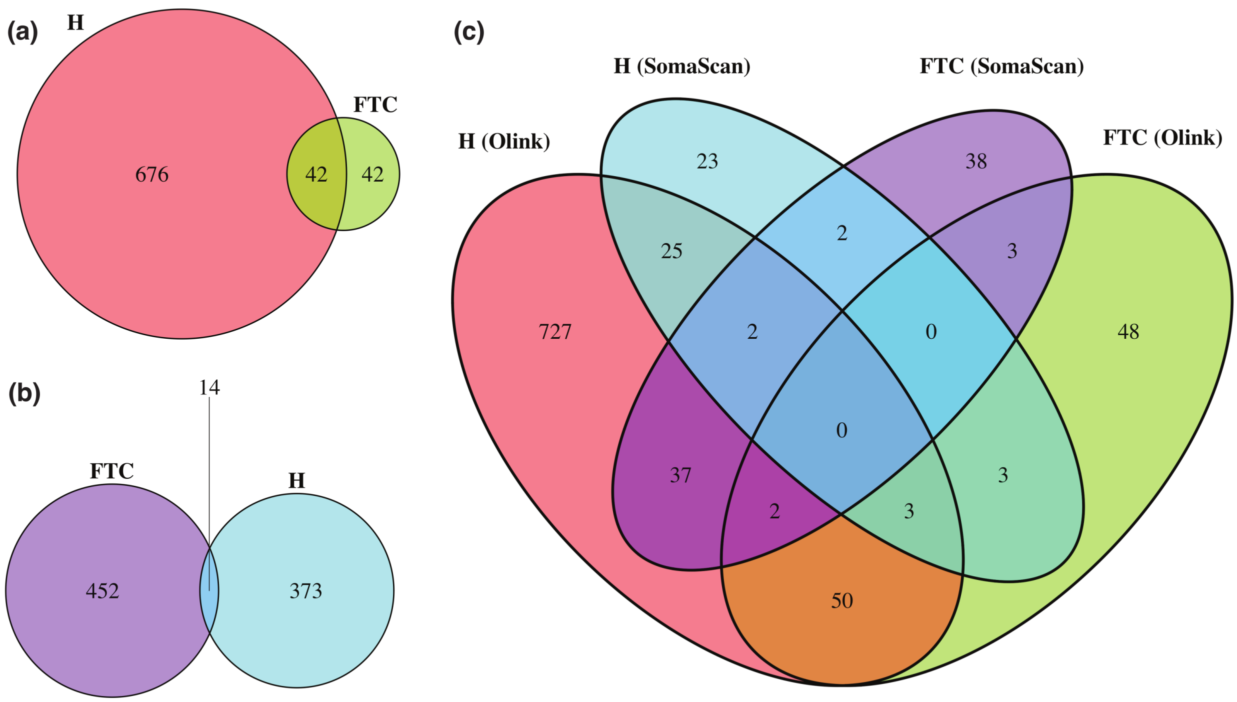

Figure 4: **Venn diagrams showing probes significantly affected by hemolysis (H) and freeze-thaw cycles (FTC). (a)** Number of affected Olink probes. **(b)** Number of affected SomaScan probes (SOMAmers). **(c)** Number of affected SOMAmer/Olink ID pairs mapped via common UniProt ID targets.

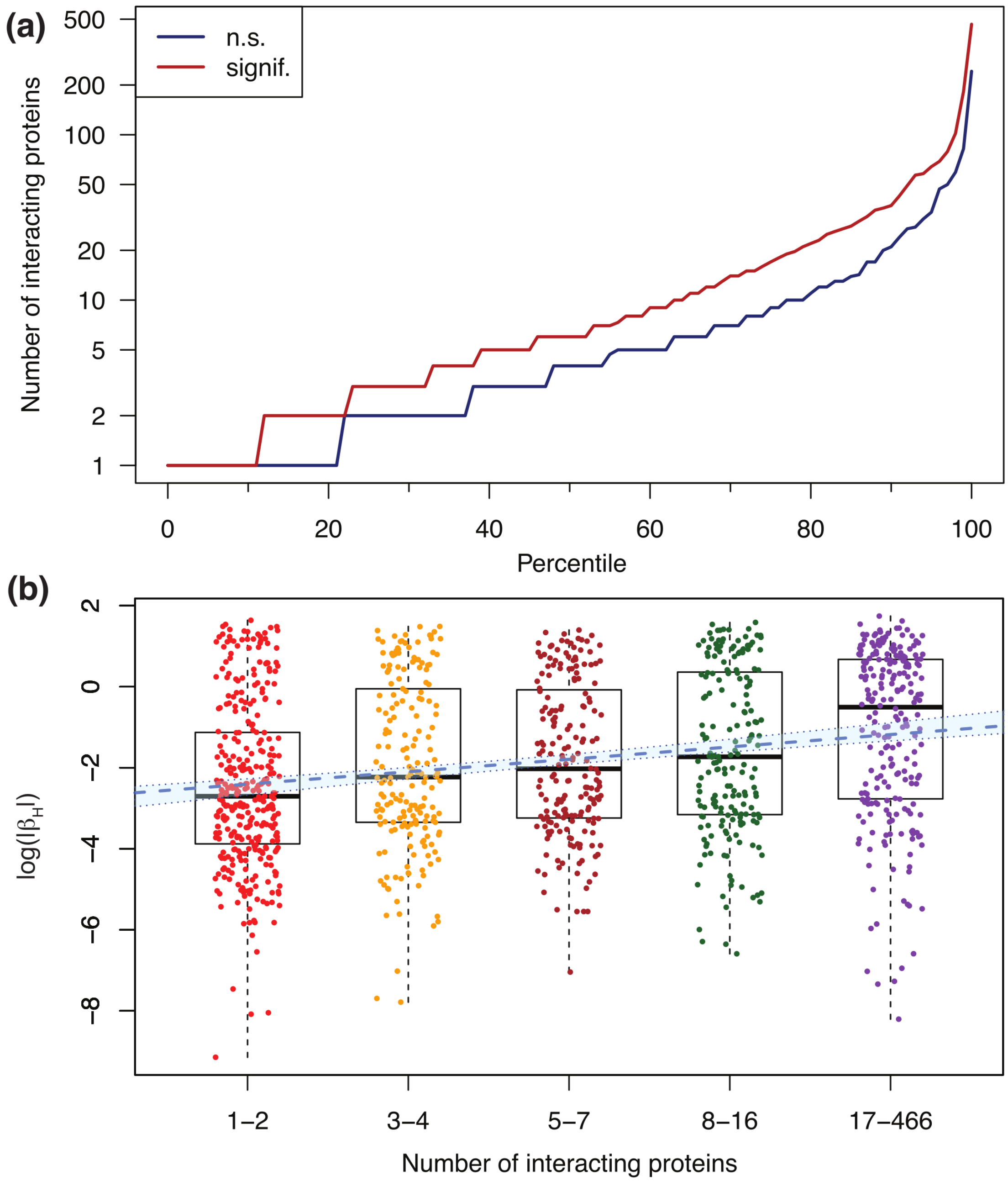

Figure 5: **Olink probes affected by hemolysis have a larger number of annotated protein-protein interactions.** **(a)** Percentile distribution comparison of the number of interacting proteins for Olink targets not significantly (n.s.) affected by hemolysis (blue) and those significantly affected by hemolysis (red). **(b)** Association between $\log(|\beta_H|)$ and the number of interacting proteins binned in five groups of similar size. Here, β_H represents the beta coefficient obtained from mixed-effects models, which captures the effect that hemolysis has on Olink's Normalized Protein eXpression (NPX) metric of abundance. The blue dashed line shows the linear regression (slope=0.31, $\mathrm{p-value} = 5.9 \times 10^{-14}$); the light-blue background enclosed within dotted boundaries shows the 95% confidence interval.